

**EN**

EM Segment Tracker

(Component not shown to scale)

1 EM Segment Tracker (9736512)

Device description

The electromagnetic (EM) Segment Trackers are navigated devices that consist of enclosed tracker coils, an implantable grade titanium screw, and a communication cable. The EM Segment Trackers are an accessory to the Stealth AXiS™ Surgical System. The EM Segment Trackers are sterile, single-use devices that are attached to vertebral segments to track anatomical movement of vertebral segments during computer-assisted surgery.

The navigation system's mobile emitter generates a low-energy magnetic field to locate the tracker coils. Then, the clinical application displays the position of vertebral segments within multiple patient image planes and other anatomical renderings.

Intended use

The EM Segment Trackers are intended to track the position of patient anatomy during spinal procedures with Medtronic computer-assisted surgery systems. The trackers are intended to be used with the Stealth AXiS™ Spine clinical application during spine procedures.

Indications for use

The Stealth AXiS™ Surgical System is intended for precise positioning of surgical instruments and as an aid for precisely locating anatomical structures in open, minimally invasive, and percutaneous procedures. The Stealth AXiS™ Surgical System is indicated for medical conditions in which the use of stereotactic surgery may be appropriate, and where reference to a rigid anatomical structure, such as the skull or vertebra, can be identified relative to a CT or MR based model, fluoroscopy images, or digitized landmarks of the anatomy.

The EM Segment Trackers are indicated for use in the thoracolumbar and sacroiliac segments.

Contraindications

None.

Material composition

For materials of concern information such as REACH, CA Prop 65, or other product stewardship programs, go to www.medtronic.com/productstewardship.

Compatibility

Segment trackers are compatible with the Stealth AXiS™ Surgical System.

System level manuals

This device can be used with Medtronic computer-assisted surgery systems. See the system level manuals and clinical application instructions for use for descriptions of patient groups, intended users, potential clinical benefits, side effects, and potential complications. This device is an accessory to your system.

See the system level manuals and the spine clinical application instructions for use for additional instructions, warnings, and precautions.

Content of this document

This document contains instructions for use for segment trackers used during spine procedures.

Warnings

Warning: Inspect sterile device packaging and expiration date before use. Do not use any pre-sterilized device whose sterile packaging appears to be compromised or whose expiration date has passed. Use of a compromised device could result in patient injury or death.

Warning: Prior to use, examine the device for damage and deterioration. Do not use any compromised device. Abandon use of any device damaged during the procedure.

Warning: Do not open sterile-barrier packages or containers until surgical use. At time of surgical use, inspect barrier for breach. If the sterile barrier is breached before surgical use, dispose of all components contained in the package.

Warning: Use caution and exercise care to prevent injury during use. See the corresponding system level manuals and clinical application instructions for use for further handling instructions.

Warning: Do not cut the segment tracker cable before unplugging it from the system. Cutting the cable could pose an electrical safety impact to the patient.

Warning: Do not hammer the segment tracker into the patient's anatomy. Hammering the tracker into bone could result in unstable fixation and damage to the tracker, which could impact navigational accuracy.

Warning: Carefully consider the location you place the segment tracker. Choose a site that fulfills these criteria:

- Will lie safely within the localizer detection field during navigation.
- Will not result in stress on the tracker because of patient positioning.
- Will not hinder access to the surgical site.
- Will not be in the immediate vicinity of any metal objects that are not designed for use with segmental tracking.
- Will remain fully seated in bone throughout the entirety of the procedure.
- Will not damage the tracker, like if placed in hard bone.
- Will not allow the tracker to move freely within bone, like if placed in soft bone.

Warning: Do not allow fluid to enter the connector portion of the device.

Warning: Post-registration movement of the patient tracker relative to its assigned level in the clinical application will result in inaccurate navigation, potentially resulting in patient harm. If you place a segment tracker after patient registration, confirm navigational accuracy by localizing on known points before you assign the segment tracker to a level.

Warning: Do not bump or reposition the segment trackers after assigning the trackers to levels in the clinical application. Movement of the segment tracker relative to the patient anatomy after assignment will result in inaccurate navigation. If the device moves in relation to the patient anatomy at any time after assignment, you must detach the segment tracker or re-register the patient with new intraoperative images.

Warning: During navigation, visually confirm navigational accuracy frequently by localizing on known points. Check the displayed navigation information and verify that the system displays the name of the tracked tool. Abort use of the system if navigation seems inaccurate and steps to restore accuracy are unsuccessful.

Warning: If you group multiple vertebral levels to one segment tracker, the clinical application will move the grouped levels according to the movement of the tracked level.

Warning: This device is for single use only. After use, dispose of the device in accordance with national regulations.

Precautions

Caution: Federal law (U.S.A) restricts this device to sale by or on the order of a physician.

Caution: Avoid damage to the segment tracker including the cable. Do not use tracked devices if there are any signs of damage, deterioration, or deformation. If you damage the device during the procedure, discontinue use of the device immediately.

Caution: Do not contact the segment trackers with active electrosurgical devices. Contacting the segment trackers with electrosurgical devices could interfere with device functionality.

Caution: Do not attempt to attach the segment tracker more than five times. After you've attempted to attach the segment tracker five times, dispose of that segment tracker and use a new one.

Use

The EM Segment Tracker is attached to vertebral segments with the EM Segment Tracker Driver to localize and to track anatomical movement between vertebral segments during spine procedures. See the Spine clinical application instructions for use for information about patient registration.

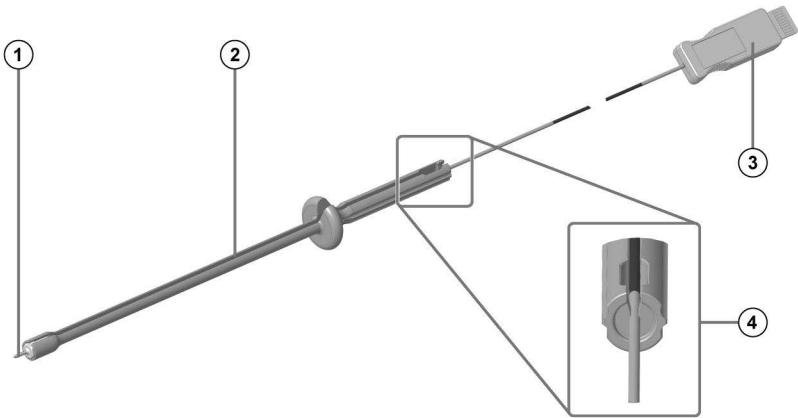
Note: When you unravel the tracker cable, consider how much of the cable needs to remain within the sterile field throughout the procedure, and then plug the connector portion of the device into the non-sterile EM instrument interface.

Note: The tracker must be placed within the navigated field produced by the emitter. For additional information about restrictions on environments in which you can use the segment trackers, see the *Emitter Quick Reference Guide*.

1. Connect the segment tracker into the EM instrument interface using the connector portion of the device. Ensure the EM instrument interface is connected to the workstation.
Note: If you want to navigate the tracker into place, complete patient registration before attaching the segment tracker to the patient.

2. Insert the segment tracker into the driver opening until the assembly is secure and fully seated.
3. Press the cable into the driver slot. Gently pull the cable taut to ensure the tracker is fully seated within the driver. Use the cable capture feature at the top of the driver secure the cable in the driver.

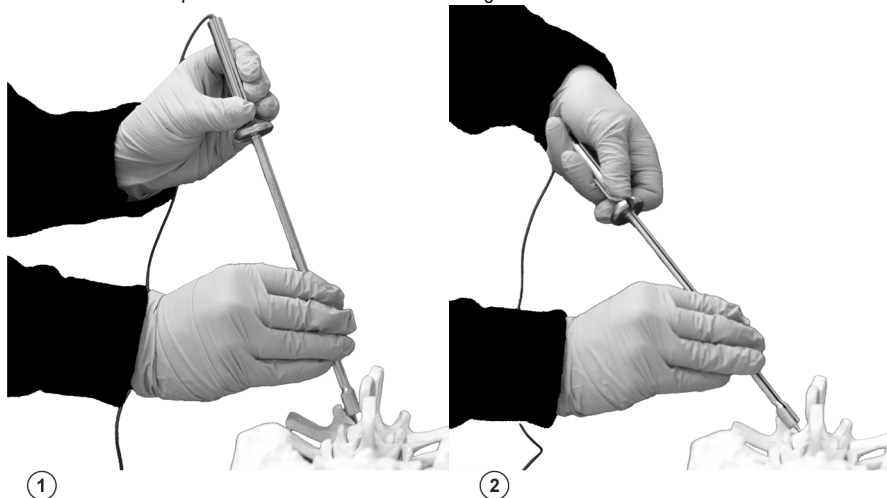
Warning: Ensure that the tracker is fully seated into the driver and that there are no particulates inside of the driver. If the tracker is not properly seated in the driver, it could damage the tracker and cause inaccurate navigation.



- 1 Segment tracker
- 2 Driver slot
- 3 Connector portion of the segment tracker
- 4 Cable capture feature

4. Attach the segment tracker into the bone using the driver.
 - a. To prevent cable wrapping, ensure that the hand turning the driver is underneath the cable and that there is enough cable in the sterile field to allow for sterile cable management throughout the procedure. Use the other hand to stabilize and guide the placement of the driver.
 - b. To minimize the chances that the tracker will slip off the bone or that you will damage the tracker, align the driver perpendicular to the bone.
 - c. Once the assembled tracker and driver are atop the desired tracker location, turn the driver clockwise while applying gentle downward force until the tracker is fully seated on bone.
 - d. If you want to track multiple vertebrae with a single tracker, consider how you want to group levels when placing a tracker on a specific level. For instructions about how to group multiple vertebral levels to one segment tracker, see the clinical application.
 - e. When attaching the EM Segment Tracker with navigation, see the *Emitter Quick Reference Guide* for emitter placement.

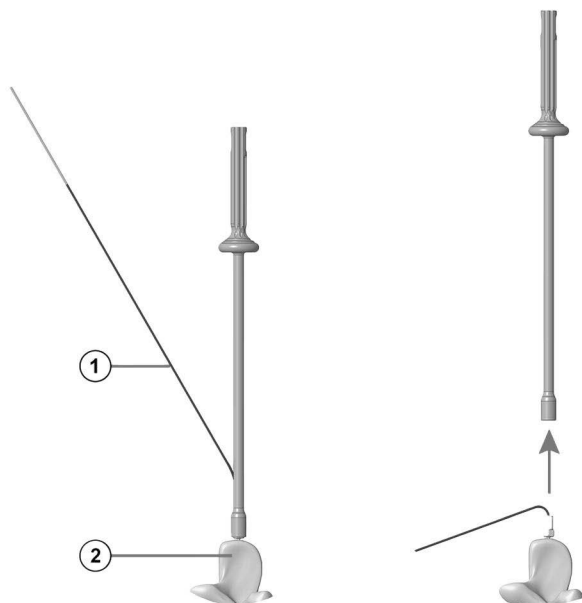
Warning: Do not apply excessive force when attaching the segment tracker to the patient. Applying excessive force could strip the bone and cause inaccurate navigation.



- 1 Hand position to gently secure tracker in bone with minimal pressure.
- 2 Hand position to apply gentle downward pressure if needed to gain purchase and progress screw into bone.

5. When the segment tracker is secured in bone, gently remove the cable from the driver slot. Gently pull up to remove the driver from the segment tracker.

Note: If you have questions about inserting the segment tracker into patient anatomy using the driver, contact Medtronic technical services.



- 1 Segment tracker cable
2 Vertebrae

6. Before you assign segment trackers, check that the trackers are secure.
 - a. Gently try to rotate the segment trackers with your fingers. If you can rotate a tracker, determine if you need to reposition or tighten it so that it is fully seated on bone.
 - b. If you are unable to manually check if the trackers are secure, see the clinical application for information about how to assign segment trackers and confirm that the tracker displays in the correct location relative to the bone model on the screen.
7. After the procedure, reattach the driver to the segment tracker and turn counterclockwise to remove the tracker from the bone.

Warning: Ensure that the tracker is fully seated into the driver and that there are no particulates inside of the driver. If the tracker is not properly seated in the driver, it could damage the tracker and cause inaccurate navigation or patient harm.

Note: Ensure the segment tracker is connected to the EM instrument interface until the segment tracker is removed from patient anatomy.
8. Dispose of the single use tracker.

Troubleshooting

Follow these steps to troubleshoot issues that occur while attaching the segment tracker:

- If the segment tracker cable wraps around the driver during insertion, verify that the cable is secured in the cable capture feature on the end of the driver and that the cable is exiting the end of the driver with a clear path. You can also have another person hold the cable exiting the end of the driver while inserting the segment tracker.
- If you use excessive force while attaching the tracker in bone, you might strip the bone. Decrease downward force on the driver and use two or three fingers to rotate the driver. Do not use your entire hand to rotate the driver.
- If the segment tracker breaks, immediately remove the broken tracker and replace it with a new tracker.
- If you have trouble successfully attaching the tracker, try inserting the tracker into a different part of the bone.
- If it feels like the segment tracker is spinning freely within the bone, attach the tracker to a different part of the bone. Do not reuse holes to insert the segment tracker as this can lead to unstable fixation and inaccurate tracking.

Note: Different bone densities may require different amounts of effort to attach the tracker to bone.

Follow these steps to troubleshoot issues that occur while using the segment tracker during navigation:

- If the segment tracker feels loose during navigation, do not try to rotate the tracker. Instead, check anatomical landmarks for accuracy. If navigation seems inaccurate, remove the tracker entirely.
- If the segment tracker or cable is damaged, bent, or fractured, immediately remove the damaged tracker and replace it with a new tracker.
- If the segment tracker stops tracking, check to see if metal objects not designed for segmental tracking is near the tracker or the emitter.
- If the segment tracker stops tracking, ensure that it is sufficiently close to the emitter.
- If tracking looks inaccurate, the segment tracker might have moved. Remove the tracker, visually inspect it for damage, reposition or replace the tracker, and reassign it.

Sterilization

Discard the device after use in accordance with the applicable regulations.

Warning: This device is for single use only. Do not reprocess, re-sterilize, or re-use this device. Attempts to reprocess or re-sterilize the device may be ineffective and may compromise its structural integrity. Any re-use creates a risk of contamination which could result in patient injury, illness, or death.

Note: This device is EO sterilized.

Disposal

Dispose of contaminated devices according to your facility's biohazard disposal procedure and local and national regulations.

Storage

Ensure the sterile packaging is dry before storing. Store in dry, clean conditions at ambient room temperature such that the package is not compromised.

Contact information

If a serious incident occurs in relation to the use of this device, report it to Medtronic Navigation. If a serious incident occurs in the European Union, also report it to the competent authority in the Member State where the incident occurred.

Assistance

If you have questions about these devices, contact your local Medtronic Navigation representative or call Medtronic Navigation, Inc. at 1 800 595 9709 or +1 720 890 3160.

Medtronic and Medtronic logo are trademarks of Medtronic.™ Third-party brands are trademarks of their respective owners. All other brands are trademarks of a Medtronic company.

This product may be protected by US Patents listed at: medtronic.com/patents

Symbol definitions

The following symbols may appear on device packaging:



Caution: Federal law (U.S.A.) restricts this device to sale by or on the order of a physician.



Consult electronic instructions for use.



Type BF applied part.

R_x Only

©2026 Medtronic Navigation, Inc.

All Rights Reserved

M059974C001DOC1

Revision A

2026-02



Medtronic Navigation, Inc.

200 Medtronic Drive

Lafayette, CO 80026

USA

Main 1 720 890 3200

Fax 1 720 890 3500

www.medtronic.com

Technical support:

USA 1 800 595 9709

International +1 720 890 3160

rs.navtechsupport@medtronic.com

www.medtronic.com/manuals

